TEAM C

Crypto Volatility and Risk Analyzer

Infosys Springboard Virtual Internship 6.0

FinTech & AI, 8 Weeks, Batch 11

Domain: FinTech & AI

Duration: 8 Weeks

Problem & Motivation

⚠️ The Problem

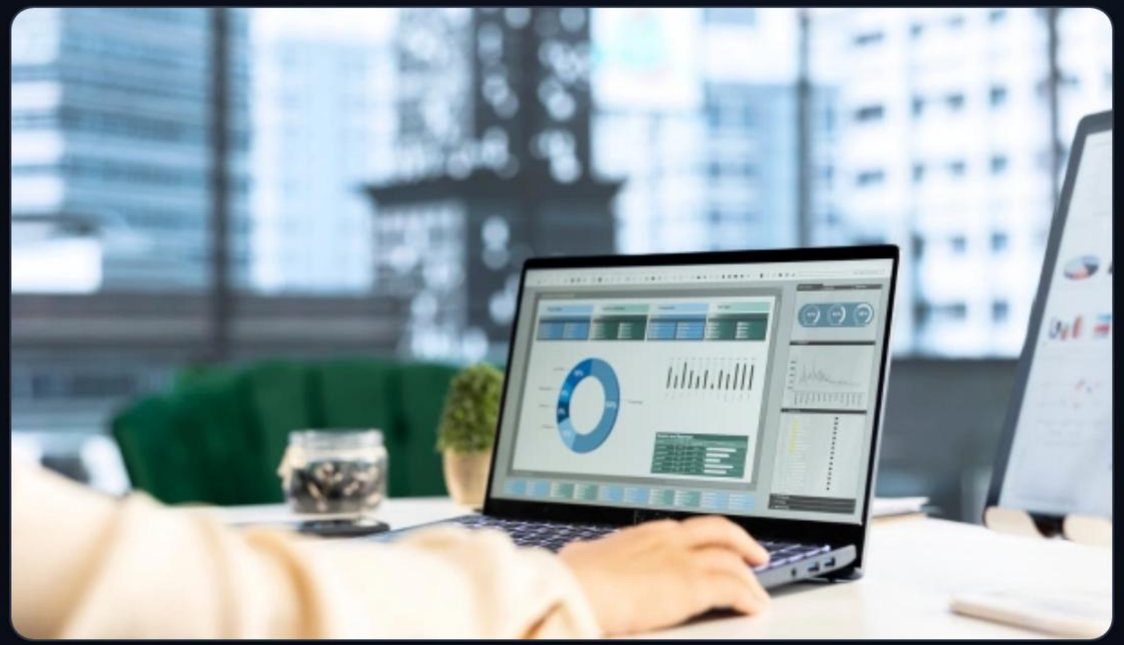
Crypto markets run 24/7 with extreme volatility, making manual risk assessment notoriously difficult for retail investors. Data is scattered across countless sources, and institutional-grade risk tools are rarely accessible to the public.

🎯 Our Goal

To democratize professional risk analytics. We aim to bridge the gap by providing a system that aggregates complex data and transforms it into understandable, actionable insights, making crypto risk transparent for beginners.

Project Objectives

- **Real-time Risk Engine:** Build a robust system to calculate risk scores on the fly for crypto assets.
- **ML-Based Clustering:** Utilize machine learning to group assets by distinct risk behaviors.
- **Explainable Insights:** Translate complex math into simple "Math to English" narratives.
- **Visual Dashboards:** Provide interactive portfolio-level risk exposure visualizations.



High-Level Architecture

1. Input Layer

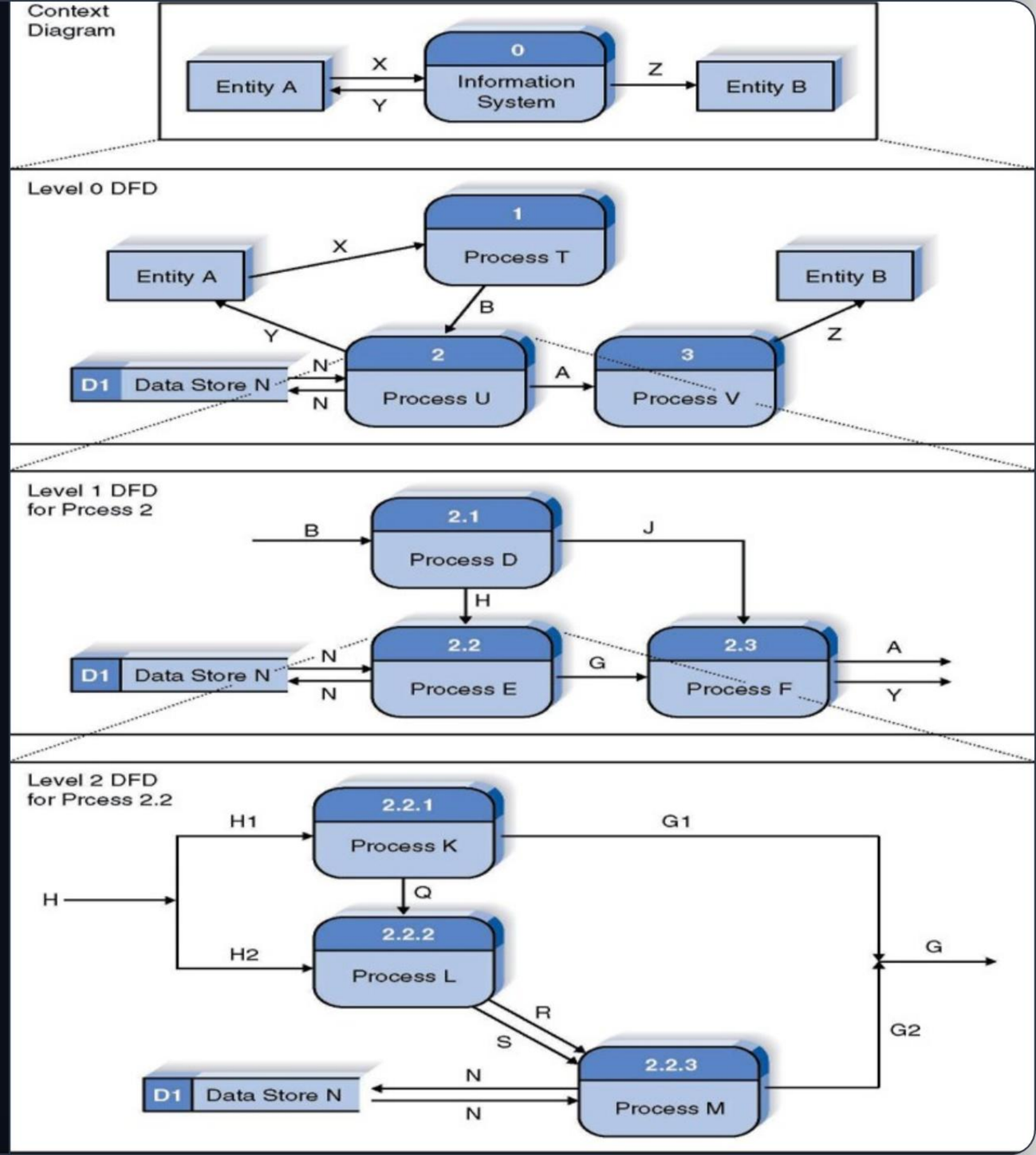
Ingests live price and volume data directly from external APIs.

2. Processing Layer

Python backend handles data cleaning, feature engineering, and Machine Learning inference.

3. Presentation Layer

A Flask web application serves interactive dashboards with a responsive UI.



Data Pipeline

1

Ingestion

Fetch historical OHLCV (Open, High, Low, Close, Volume) data via APIs.

2

Cleaning

Handle missing values and rigorously remove data anomalies and outliers.

3

Normalization

Apply Min-Max scaling to ensure consistent inputs for ML models.

4

Windowing

Create rolling windows (e.g., 7-day, 30-day) to capture temporal trends.

Core Risk Metrics (Existing)

Rolling Volatility

Measures the instability of returns over a specific time window. Higher values indicate a more unstable asset.

Maximum Drawdown

The largest peak-to-trough fall. It approximates the worst-case loss scenario for an investor.

Value at Risk (VaR)

Estimates the maximum expected loss over a defined period at a given confidence level (e.g., 95%).

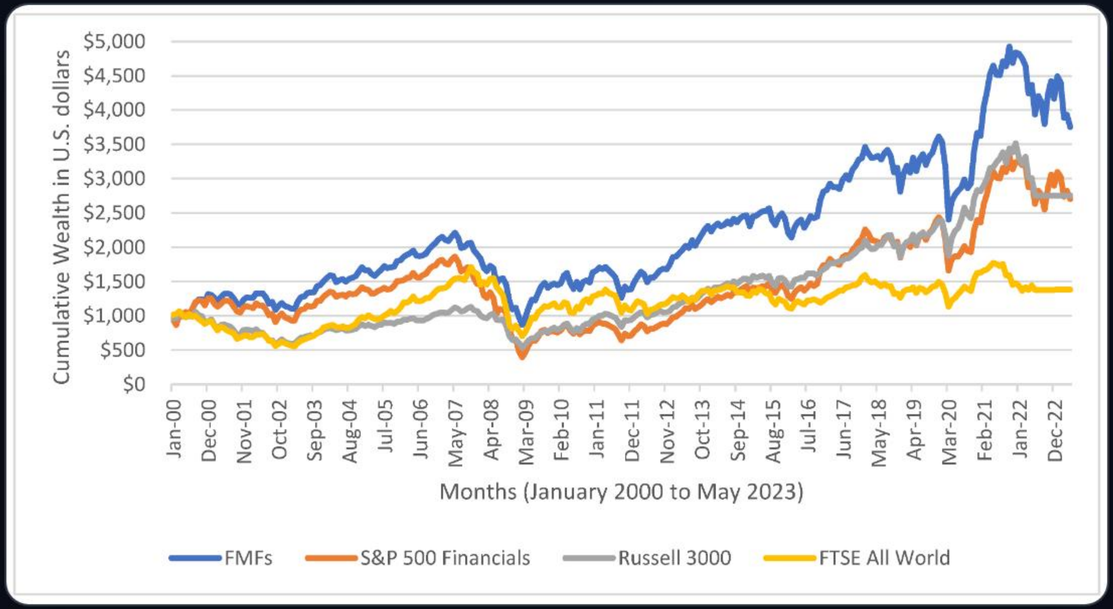
Risk-Adjusted Metrics (New)

Efficiency Over Raw Returns

We analyze how efficiently returns are earned relative to the risk taken.

- **Sharpe Ratio:** Return per unit of *total* volatility. Useful for general efficiency.
- **Sortino Ratio:** Return per unit of *downside* volatility. Focuses only on harmful risk, making it ideal for crypto.

> **Use Case:** Compare two coins not just on returns, but on risk-adjusted performance.

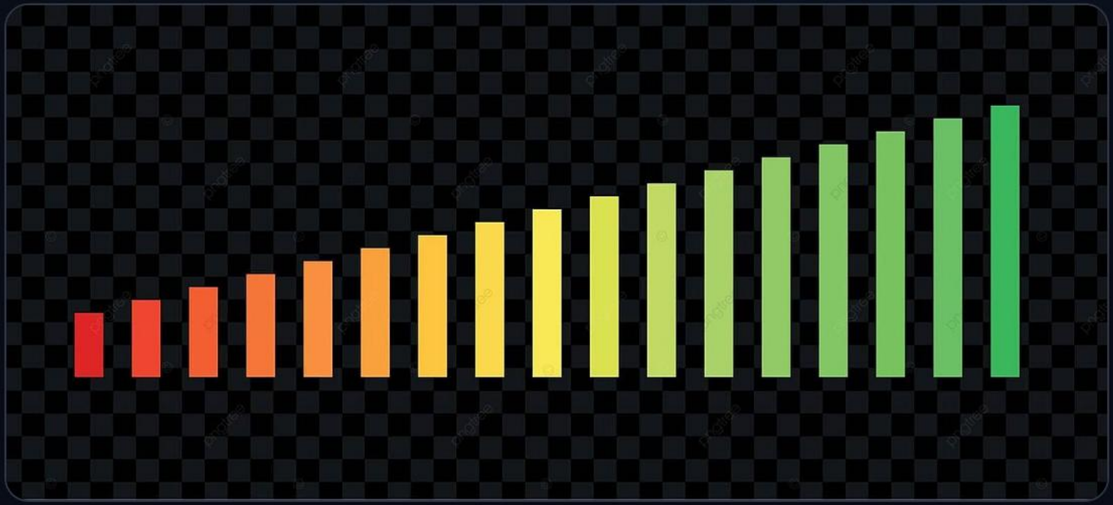


Risk Scoring & Asset Categories

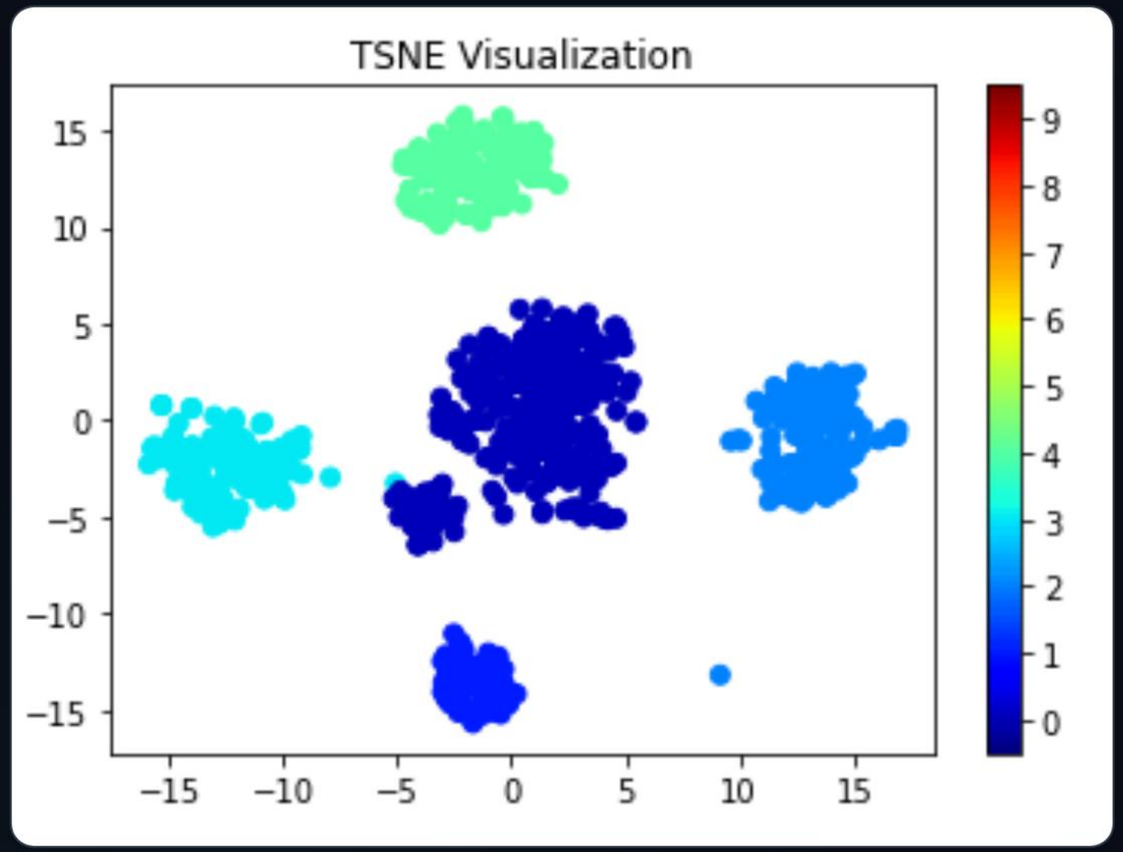
Normalized Risk Score (0–100)

We combine volatility, drawdown, VaR, and risk-adjusted metrics into a single, easy-to-understand score.

- **0-30** **Low Risk:** Stablecoins, relatively stable assets.
- **31-70** **Medium Risk:** Major coins like Bitcoin, Ethereum.
- **71-100** **High Risk:** Speculative altcoins & meme tokens.



ML Clustering: Behavior-Based Groups



K-Means Clustering

Machine learning groups assets based on actual market behavior (volatility, drawdown, volume) rather than just labels.

- **Visual Clusters:** The plot shows distinct groups of low, medium, and high-risk assets.
- **Behavioral Insight:** Helps users identify which coins "behave similarly" in the market, aiding in diversification.

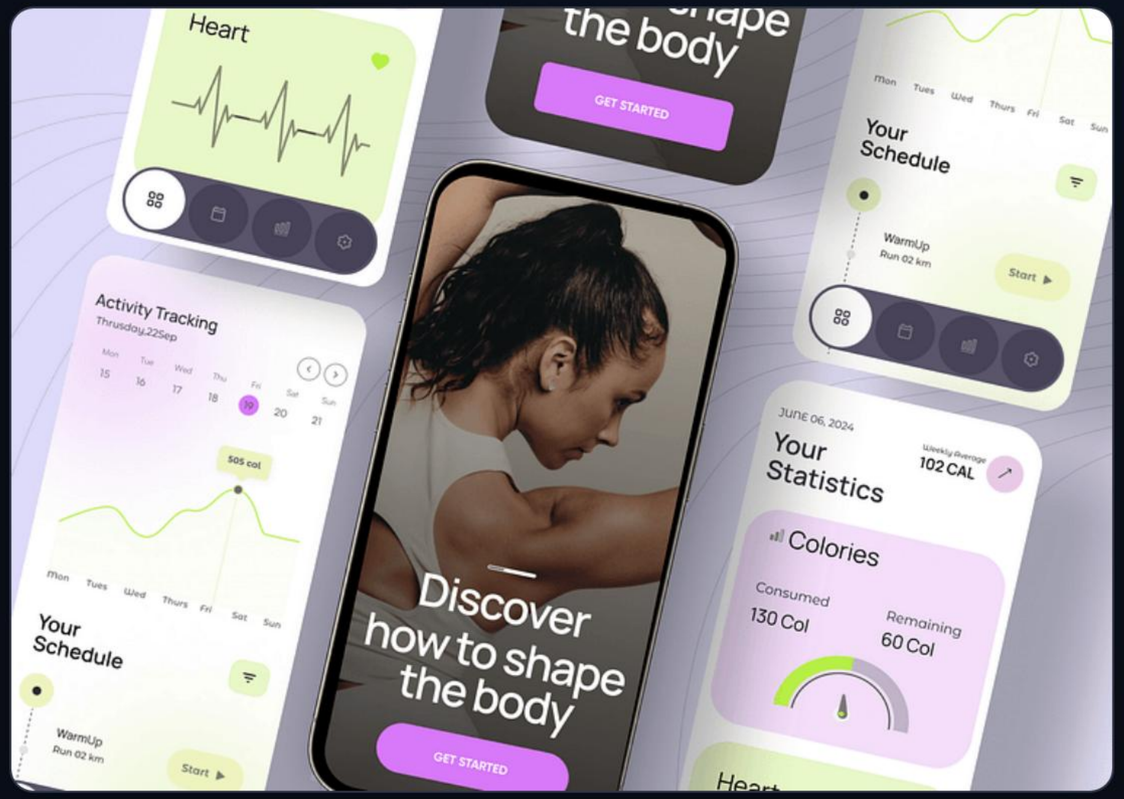
Insight Explanation Engine

”Math to English”

Our core innovation converts complex statistical metrics into plain-English insights, making risk understandable for everyone.

```
Raw Data: Vol_30d: 0.85, MaxDD: -45%
```

Insight: "This coin has been very unstable and recently fell 45% from its peak. It is very risky for short-term traders."



Time Horizon Analysis

Short-Term (7 Days)

Focus: News events, liquidity shocks, and short-term volatility spikes.

Use Case: Active traders deciding entry/exit timing.

Key Metric: Hourly/Short-term Volatility

Long-Term (90+ Days)

Focus: Adoption trends, macro factors, and drawdown recovery patterns.

Use Case: Long-term holders and portfolio builders.

Key Metric: Maximum Drawdown & Recovery

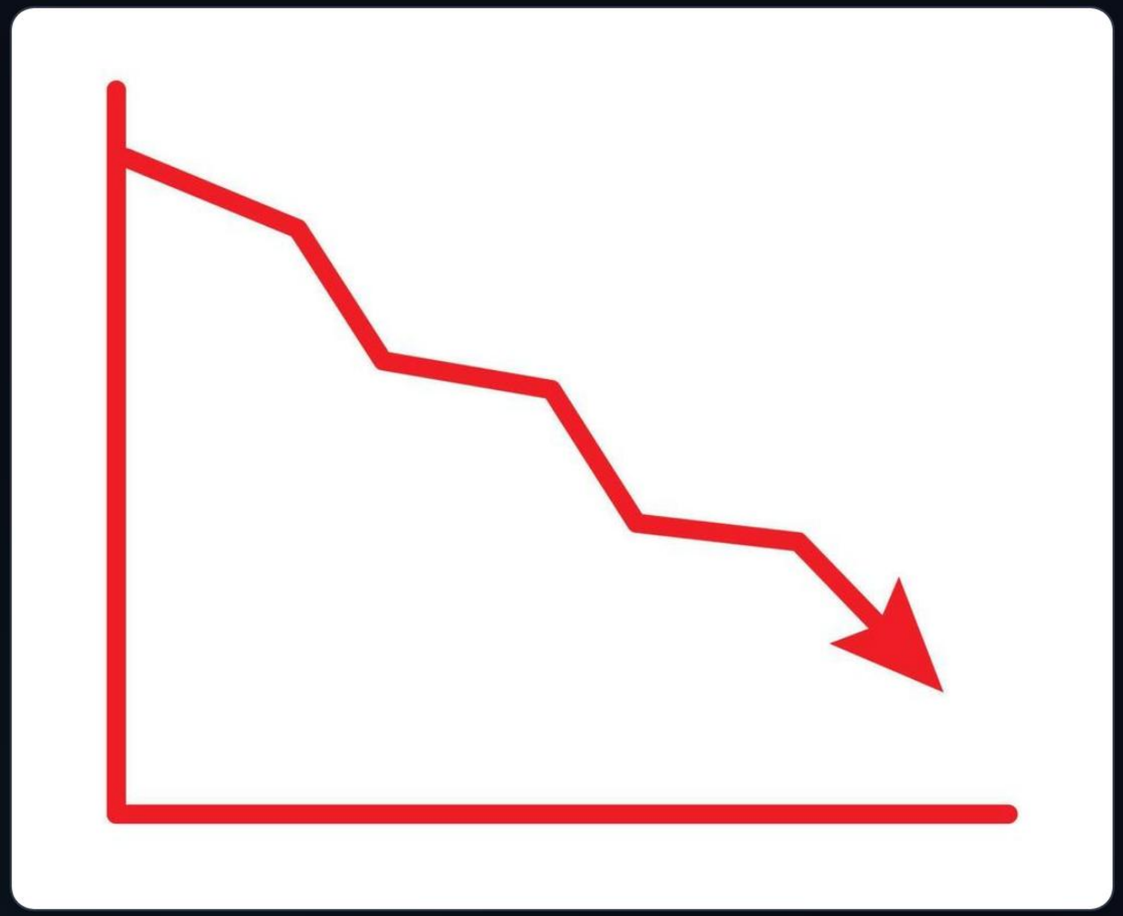
Portfolio & Correlation Analysis

Understanding Connections

- **Correlation Matrix:** Shows which assets move together. High correlation reduces diversification benefits.
- **Diversification Score (0–10):** Indicates how well-balanced the portfolio is against common market shocks.
- **Risk Aggregation:** A weighted average risk based on position sizes, highlighting which assets contribute most to total risk.

	BTC	ETH	BNB	AVAX	ADA	MATIC	WOO	SOL	DOT	SPX
BTC	1	0.719	0.499	0.378	0.434	0.657	0.64	0.539	0.488	0.057
ETH	0.719	1	0.18	0.347	0.284	0.381	0.219	0.633	0.661	0.298
BNB	0.499	0.18	1	-0.398	-0.104	0.424	0.462	-0.23	0.234	-0.351
AVAX	0.378	0.347	-0.398	1	0.835	0.543	0.469	0.875	0.195	0.358
ADA	0.434	0.284	-0.104	0.835	1	0.716	0.664	0.793	0.291	0.283
MATIC	0.657	0.381	0.424	0.543	0.716	1	0.861	0.56	0.346	-0.111
WOO	0.64	0.219	0.462	0.469	0.664	0.861	1	0.401	0.258	-0.301
SOL	0.539	0.633	-0.23	0.875	0.793	0.56	0.401	1	0.404	0.549
DOT	0.488	0.661	0.234	0.195	0.291	0.346	0.258	0.404	1	0.203
SPX	0.057	0.298	-0.351	0.358	0.283	-0.111	-0.301	0.549	0.203	1

Stress Testing & "What-If" Scenarios



Simulating Market Conditions

We simulate how a portfolio behaves under different stress conditions to test resilience.

- **Market Correction:** -20% drop across the market.
- **Crypto Winter:** -70% deep drawdown.
- **Liquidity Shock:** Sudden volume drop & spread widening.
- **Regulatory Shock:** Major coin delisted or blocked.

Web Application & Tech Stack

Built for Accessibility

- **Web Interface:** Users access the system via a Flask web app featuring a responsive UI and interactive charts.
- **Backend & ML:** A powerful Python stack handles data processing, metrics computation, clustering, and insight generation.
- **UX Design:** Designed to be clean and intuitive, allowing non-technical crypto beginners to navigate with ease.

Outcomes, Limitations & Future Scope

Key Outcomes

- **Educational:** Users learn risk beyond price.
- **Decision Support:** Encourages data-driven choices.
- **Transparency:** White-box metrics build trust.

Future Scope

- Add Sentiment Analysis (NLP).
- Real-time rebalancing suggestions.
- DeFi and liquidity pool risk scoring.

Image Sources

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